

Predicting Violent Crime Hotspots in Chicago Through Urban Disorder Indicators

Violent crime in Chicago is not randomly distributed. It clusters in specific census tracts and shifts over time in ways that correlate with observable neighborhood conditions. This section presents the results of four regression models designed to measure the relationship between urban disorder indicators and violent crime rates across 789 census tracts from 2018 to 2025.

The first model, a pooled ordinary least squares (OLS) regression with year fixed effects, tested whether tracts with higher levels of disorder also tend to have higher violent crime rates. The dependent variable was the log-transformed violent crime rate per 1,000 residents, and the independent variables were eight log-transformed disorder indicators: graffiti complaints, lighting complaints, infrastructure complaints, abandonment and building violations, rat and rodent complaints, sanitation code violations, property and street crime, and vice and public disorder. The model also controlled for population size and common year-level shocks. This specification explained approximately 79 percent of the variation in violent crime rates across the panel ($R^2 = 0.788$). Nearly all disorder indicators were statistically significant at the one percent level, with property and street crime (including theft, criminal damage, burglary, motor vehicle theft, and weapons violations; coefficient = 0.597) and vice and public disorder (including narcotics, prostitution, public peace violations, liquor law violations, and gambling; coefficient = 0.275) showing the largest positive associations. A composite disorder index averaging all eight indicators was also tested and found to be highly significant, confirming that overall neighborhood disorder is strongly associated with higher violent crime (*see Figure 2*).

The second model disaggregated the outcome variable into five specific violent crime categories: assault, battery, robbery, homicide, and criminal sexual assault. Property and street crime was the strongest predictor across all five categories, with the largest coefficient observed for robbery (coefficient = 0.673, $p < 0.01$). Abandonment and building violations showed consistently positive associations across all five crime types, and vice and public disorder were also significant for every category (*see Figure 1*). Infrastructure complaints, by contrast, were consistently negative, which may reflect differences in reporting patterns rather than a direct relationship with crime.

While the first two models demonstrated strong cross-sectional associations, they could not distinguish whether disorder indicators are genuinely linked to violent crime or whether the correlation simply reflects the fact that some neighborhoods are persistently disadvantaged. To address this, the third model introduced tract-level fixed effects alongside year fixed effects, effectively comparing each neighborhood only to itself over time. This is a substantially stronger test: rather than asking whether high-disorder neighborhoods have more crime, it asks whether a given neighborhood experiences more violent crime in years when its disorder indicators increase. After adding these controls, the model's overall R^2 rose to 0.924, but most disorder indicators lost statistical significance. Graffiti, infrastructure complaints, abandonment violations, and sanitation code violations all became insignificant, indicating that their earlier correlations were driven by persistent neighborhood characteristics rather than year-to-year

changes. Only property and street crime (coefficient = 0.236, $p < 0.01$) and vice and public disorder (coefficient = 0.039, $p < 0.01$) remained robust (*see Figure 3*).

The fourth model tested the predictive dimension directly by lagging all disorder indicators by one year. Instead of examining whether disorder and crime move together in the same year, this model asked whether last year's disorder levels predict this year's violent crime rate. Under the tract-plus-year fixed effects specification, lagged property and street crime were the strongest predictors (coefficient = 0.107, $p < 0.01$), and lagged vice and public disorder were also significant (coefficient = 0.042, $p < 0.01$). These results indicate that within a given neighborhood, a rise in property crime or public disorder in one year is followed by an increase in violent crime the next year. This pattern may reflect that property and street crime captures actual criminal activity rather than complaint-based reporting, making it less sensitive to variation in reporting behavior and more directly tied to underlying dynamics of crime escalation (*see Figure 4*).

To further test the robustness of these findings, the lagged specification was also disaggregated by crime type, mirroring the structure of the second model but with all indicators shifted back one year. This analysis revealed important differences from the contemporaneous results. Once lagged, most disorder indicators that had appeared significant in the cross-sectional model lost their predictive power, consistent with the fixed effects findings. However, lagged property and street crime remained the dominant predictors across assault, battery, and robbery, with a particularly strong model fit for assault and battery ($R^2 \approx 0.85$). Abandonment and building violations also retained significance in the lagged specification, suggesting that vacant and neglected properties may act as a slower-developing but genuine early warning sign (*see Figure 5*). Criminal sexual assault proved largely unpredictable across all specifications, likely due to severe underreporting that limits the data's ability to capture meaningful variation.

Spatial analysis using ArcGIS reinforced these findings. Choropleth maps of homicide counts and vacant building violations by census tract revealed striking visual overlap, with the highest concentrations in both maps clustered on Chicago's south and west sides (*see Figures 7 and 8*). This pattern is not limited to homicide. Maps of assault, battery, and robbery show nearly identical spatial concentration in the same set of tracts, indicating that multiple forms of violent crime are co-located in these areas (*see Figures 9–11*). Criminal sexual assault follows a similar but more diffuse pattern, which is consistent with its weaker statistical results and known issues with underreporting (*see Figure 12*). These maps demonstrate that the concentration of violent crime is not driven by a single category but reflects broader, persistent geographic clustering. The same areas that exhibit high levels of disorder and abandonment are also the areas where multiple types of violent crime occur at elevated rates. This spatial consistency strengthens the regression findings and provides clear visual evidence supporting geographically targeted interventions.

As a supplemental test of external validity, the same lagged regression framework was applied to crime data from Los Angeles, covering 1,212 census tracts from 2020 to 2024 (4,651 observations after lagging). The pooled OLS results mirrored the Chicago findings: lagged property and street crime were the strongest predictors of current-year violent crime (coefficient = 0.337, $p < 0.01$), followed by graffiti (0.156, $p < 0.01$) and vice and public disorder (0.058, $p < 0.05$). However, after adding tract and year fixed effects, none of the lagged indicators retained statistical significance. This pattern is consistent with the Chicago results in an important way: it

confirms that disorder indicators are strongly correlated with violent crime across neighborhoods, making them useful for identifying high-risk areas, but the within-tract predictive signal is weaker in Los Angeles, possibly due to the shorter panel (four years versus seven) or differences in how the two cities collect and categorize disorder complaints. The LA results reinforce the practical value of these indicators for flagging at-risk neighborhoods while underscoring that the causal interpretation should remain cautious.

Several factors may confound the relationships observed in the regression results and threaten the validity of the analysis. The contrast between the pooled and fixed effects models indicates that many observed relationships are driven by persistent differences across neighborhoods rather than changes within them over time. This suggests that variables such as long-term economic disadvantage, housing stability, and structural disinvestment are influencing both disorder indicators and violent crime simultaneously. These underlying conditions are not directly observed in the model but are embedded in the baseline differences across tracts, which inflate cross-sectional correlations. In addition, reverse causality may be present in several indicators. Increases in violent crime can lead to higher reporting of disorder, greater police presence, and more recorded minor offenses, making it difficult to fully separate cause from effect even in lagged specifications. These relationships may also be influenced by omitted variables not directly included in the model, such as income levels, employment conditions, and education, all of which may affect both disorder indicators and violent crime.

Measurement-related confounders further complicate interpretation. Many of the independent variables are derived from 311 service requests, which are not purely objective measures of disorder. Reporting rates depend on resident engagement, awareness, and access to city services, all of which vary systematically across neighborhoods. As a result, higher 311 activity may reflect stronger reporting behavior rather than higher underlying disorder. Similarly, recorded crime data may be influenced by differences in policing intensity and deployment. Areas with greater police presence are more likely to detect and record both minor offenses and violent crime, which can artificially strengthen observed relationships between indicators. This issue is particularly relevant for categories such as narcotics, public disorder, and minor theft, which are more sensitive to enforcement patterns. In addition, certain dependent variables suffer from structural measurement limitations. Criminal sexual assault, for example, is widely underreported, which reduces variation in the data and explains its weak and inconsistent performance across all model specifications. In addition, the construction of composite indicators and the grouping of categories introduces a degree of researcher-driven subjectivity. Alternative classification choices or variable definitions could produce different results, particularly for broader categories such as disorder and infrastructure complaints.

Temporal and structural factors also introduce potential bias. The datasets combine event-based measures such as crime incidents and 311 complaints with slower-moving structural variables such as population and housing conditions. Misalignment in timing can distort relationships if indicators and outcomes are not measured over consistent intervals. Seasonality, citywide shocks, and major events may simultaneously affect both disorder indicators and crime rates, creating spurious correlations in models that do not adequately control for time effects. Additionally, spatial aggregation introduces its own limitations. Census tracts are used as the unit of analysis, but crime and disorder do not conform to these boundaries. Activity in one tract may

influence neighboring areas, creating spatial spillover effects that are not fully captured in a tract-level model and may bias estimated relationships.

Several steps were taken to mitigate these confounding influences and strengthen the validity of the results. First, violent crime was normalized by population to produce rates rather than raw counts, reducing bias driven by differences in tract size. All datasets were aligned to a consistent geographic unit and standardized to annual time intervals to ensure comparability across sources. The inclusion of tract-level fixed effects serves as the primary safeguard against confounding by controlling for all time-invariant neighborhood characteristics, including those not directly observed in the data. Year fixed effects further control for citywide shocks, seasonal patterns, and macro-level changes that affect all tracts simultaneously.

To address issues of simultaneity and reverse causality, lagged versions of all disorder indicators were introduced. This shifts the analysis from contemporaneous correlation to temporal ordering, testing whether increases in disorder precede increases in violent crime rather than occur alongside them. The fact that only a small subset of indicators remains significant under lagged fixed effects strengthens confidence that these relationships are not solely driven by concurrent movement. Additional robustness is provided by disaggregating the dependent variable into specific crime types and by replicating the lagged framework in Los Angeles. The LA results follow the same general pattern, reinforcing the interpretation that many indicators capture stable neighborhood differences rather than causal drivers of change.

Despite these mitigation strategies, some limitations remain. Reporting bias in both crime and 311 data cannot be fully eliminated, and time-varying factors such as changes in policing strategy, economic shocks, or policy interventions may still influence the results. The relatively short time horizon of both datasets, particularly in Los Angeles, limits the ability to capture longer-term structural dynamics and may weaken the detection of delayed effects. The findings should therefore be interpreted as identifying consistent predictive relationships rather than definitive causal mechanisms.

Taken together, these results carry a clear policy implication. Not all disorder indicators are equally informative. Only property and street crime complaints, vice and public disorder requests, and, to a lesser extent, abandonment and building violations reliably signal that a neighborhood is trending toward higher violence in the following year. This analysis advocates for the City of Chicago to act on that finding by formalizing a proactive, data-driven resource allocation strategy, one that monitors within-tract changes in these specific indicators and directs preventive interventions toward high-risk neighborhoods before violent crime fully escalates.

The expected net benefit of this approach is favorable. Cross-city data from Los Angeles reinforces this finding: census tracts with the highest property and street crime complaint volumes averaged 29.1 violent crimes per 1,000 residents, more than three times the rate of the lowest three quartiles (*see Figure 6*). Economists estimate that each homicide carries a minimum social cost of approximately \$1.5 million when accounting for medical care, lost productivity, and criminal justice expenditures - a figure that does not capture the full downstream burden of reduced property values, business disinvestment, and lost employment opportunities that compound over time in affected neighborhoods (Khubchandani & Price, 2017). When the full cost of violence is considered across assault, battery, robbery, and criminal sexual assault in addition to homicide, the aggregate annual economic burden on affected neighborhoods runs into

the billions. Drawing on these estimates and applying them to the tract-level crime rates identified in our analysis, the expected net benefit of early intervention is favorable. Given that the regression results establish association rather than causation, the authors present this as a directional net benefit assessment rather than a precise cost-benefit calculation. The available evidence consistently points toward favorable returns on early intervention, but the exact magnitude will depend on implementation scale and local conditions.

The interventions most directly supported by the data are low-cost relative to the harm they aim to prevent. Abandonment and building violations, which retained significance as a lagged predictor of homicide even after controlling for fixed neighborhood characteristics, are a natural first target. Research on abandoned building remediation has identified securing vacant structures as a relatively low-cost method of reducing crime in and around those properties. Studies of vacant lot remediation programs estimate that \$1 invested in cleanup and greening returns approximately \$26 in net benefits to taxpayers from reductions in gun violence alone (Manhattan Institute, 2023). Accelerating building code enforcement and remediation in tracts where abandonment complaints are rising relative to their historical baseline is therefore among the highest-return investments the city can make with limited prevention resources.

Environmental remediation alone, however, is not sufficient to address the two indicators that proved most robust across all model specifications. Property and street crime complaints and vice and public disorder requests reflect social and behavioral dynamics that physical cleanup cannot fully resolve. For these, the evidence points toward community violence intervention as a necessary complement. A peer-reviewed quasi-experimental study of community violence intervention programs found that program alumni were 73.4% less likely to have an arrest for a violent crime in the two years following enrollment, a reduction that is large given the complex systemic factors contributing to gun violence (Papachristos et al., 2023). The Johns Hopkins Center for Gun Violence Solutions similarly identifies CVI programs as among the most evidence-supported tools available for reducing community-level violence, emphasizing street outreach by credible messengers and behavioral science-informed approaches as core components (Johns Hopkins Center for Gun Violence Solutions, n.d.). Chicago's own CVI infrastructure has already demonstrated results, with a recent city report noting a measurable reduction in retaliatory violence and individuals who completed the program being 73% less likely to be arrested for a violent crime in the two years following enrollment (City of Chicago, n.d.). Directing CVI resources toward the same tracts flagged by the disorder indicator framework, those showing year-over-year spikes in property and street crime and public disorder relative to their own baseline, would create a layered strategy that addresses both environmental and social dimensions of violence in the same locations at the same time.

The practical implementation of this recommendation does not require the creation of new programs or institutions. The 311 system already captures real-time, tract-level disorder complaint data. The Department of Buildings already has enforcement authority over vacant properties. CVI organizations already operate in Chicago's highest-violence communities. What is needed is a systematic process for using the regression framework developed in this analysis to connect those existing resources to the specific tracts where the indicators are trending in the wrong direction. Chicago's People's Plan for Community Safety has already demonstrated the value of this kind of data-driven, hyperlocal approach, using quantitative and qualitative analysis to prioritize specific block groups for targeted investment based on violence indicators and

measures of historical disinvestment (City of Chicago, n.d.). Extending that logic citywide using within-tract changes in property crime, public disorder, and abandonment complaints as the trigger for resource deployment would make that framework both more systematic and more scalable.

It is important to be clear about the limitations of this recommendation. The regression results are correlational, and the proposed interventions are grounded in expected net benefit rather than proven causal impact. The analysis cannot definitively establish that disorder indicators cause violent crime, only that they are consistently associated with it across multiple model specifications, crime types, and locations. That said, the consistency of the findings, particularly the survival of property and street crime and vice and public disorder through the strictest fixed-effects and lagged specifications combined with the strong external evidence on the cost-effectiveness of environmental remediation and community violence intervention, provides a reasonable and defensible basis for action. A city that waits for causal proof before committing resources accepts the ongoing costs of violence in the interim. An evidence-informed, proactive approach that uses available data to identify where conditions are worsening and acts on those signals early, represents a more efficient and defensible allocation of public resources than one organized entirely around reactive response.

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Appendix

Figure 1 - Regression Coefficients: Disorder Indicators vs. Violent Crime Types

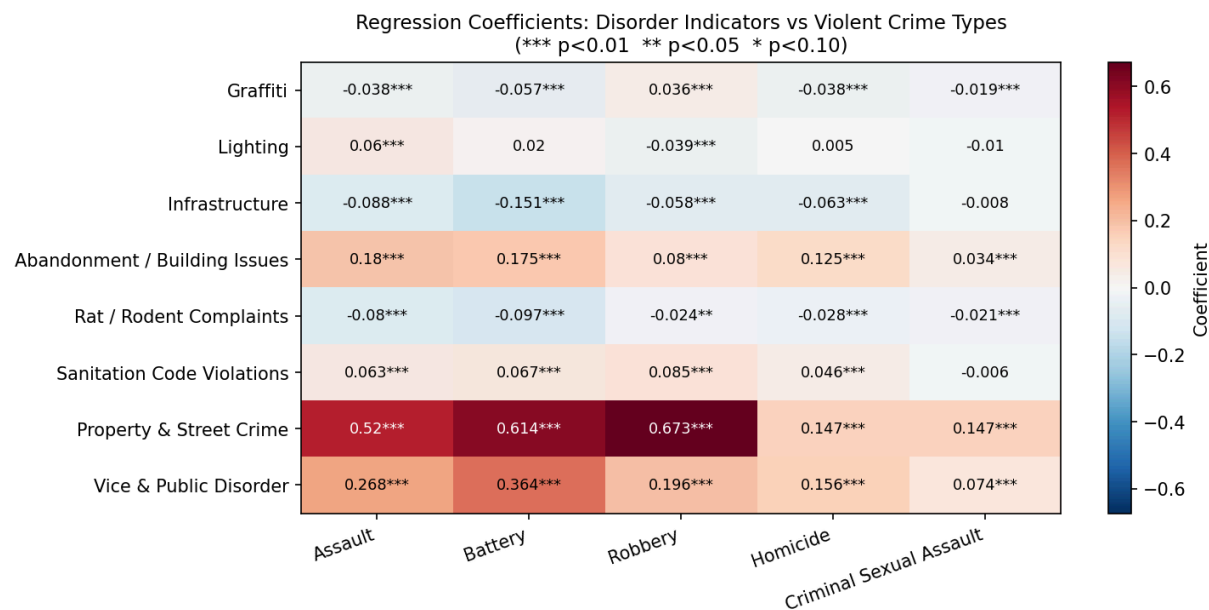


Figure 2 - Pooled OLS Effects of Disorder Indicators on Violent Crime Rate (95% Confidence Interval)

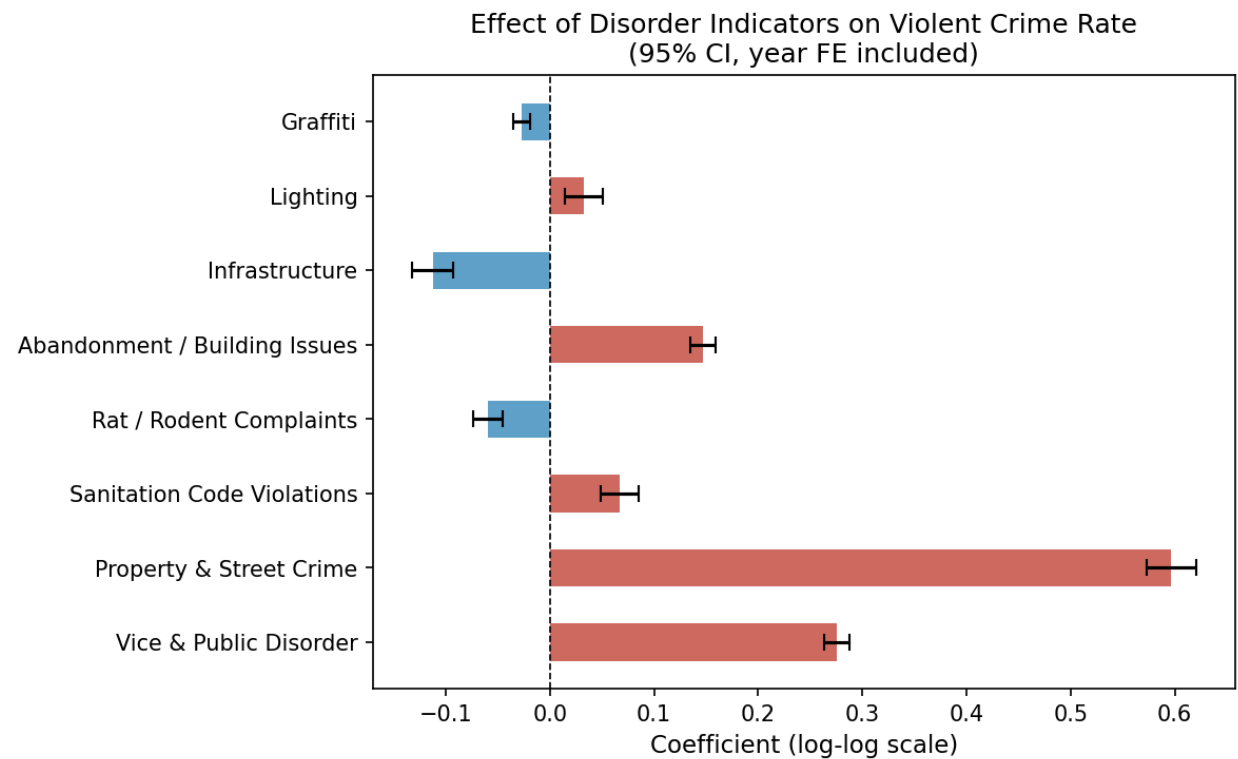


Figure 3 - Effect of Disorder Indicators on Violent Crime Rate: Pooled OLS vs. Fixed Effects Models

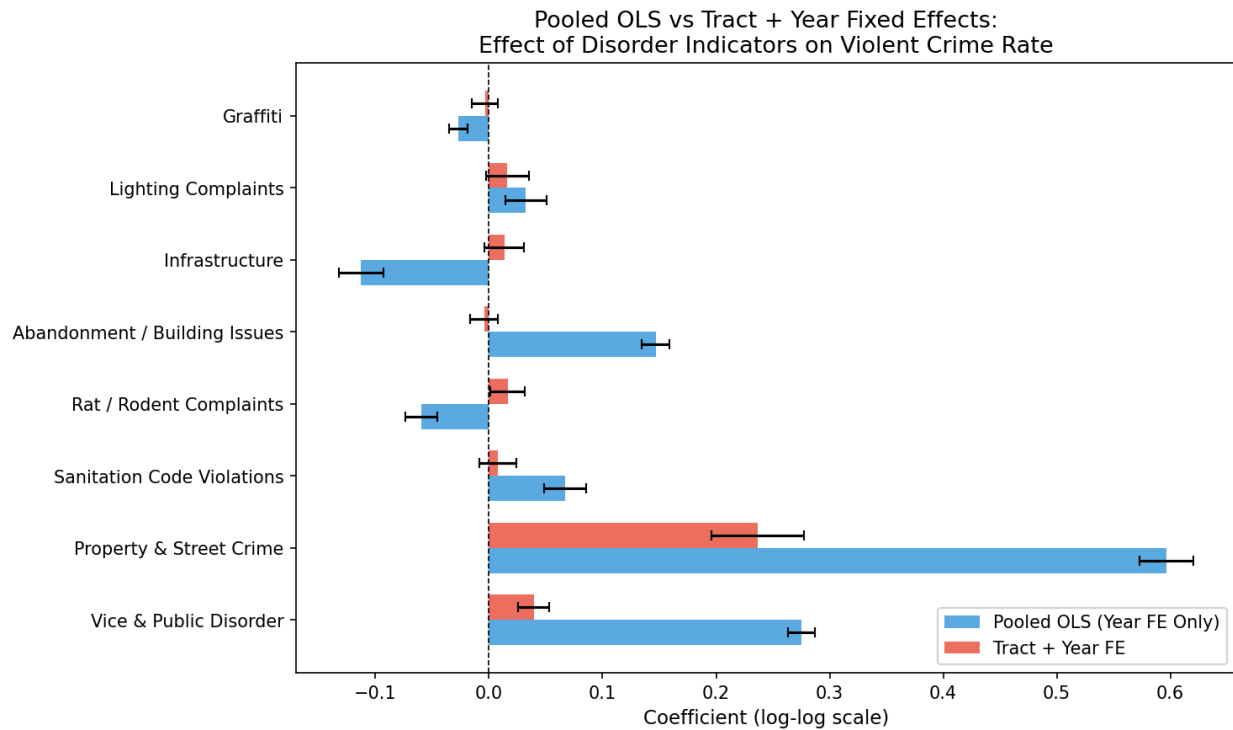


Figure 4 - Lagged Effects of Disorder Indicators on Next-Year Violent Crime Rate

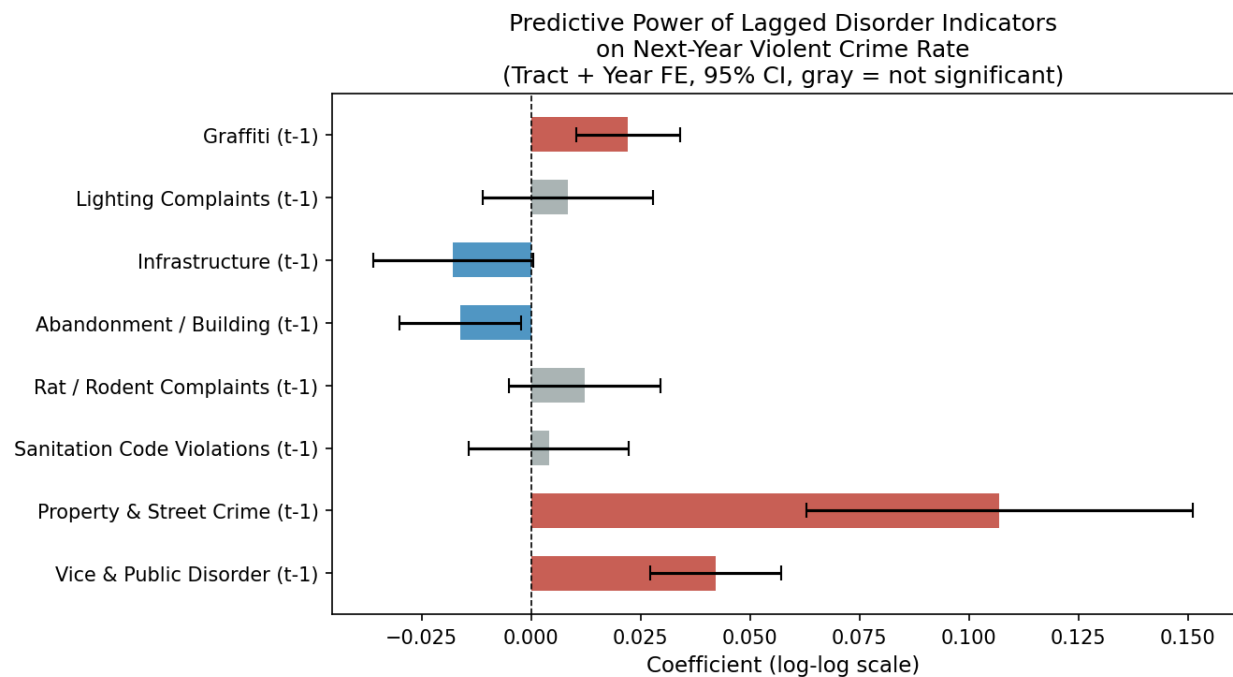


Figure 5 - Lagged Regression Coefficients Matrix (Disorder Indicators vs. Violent Crime Types)

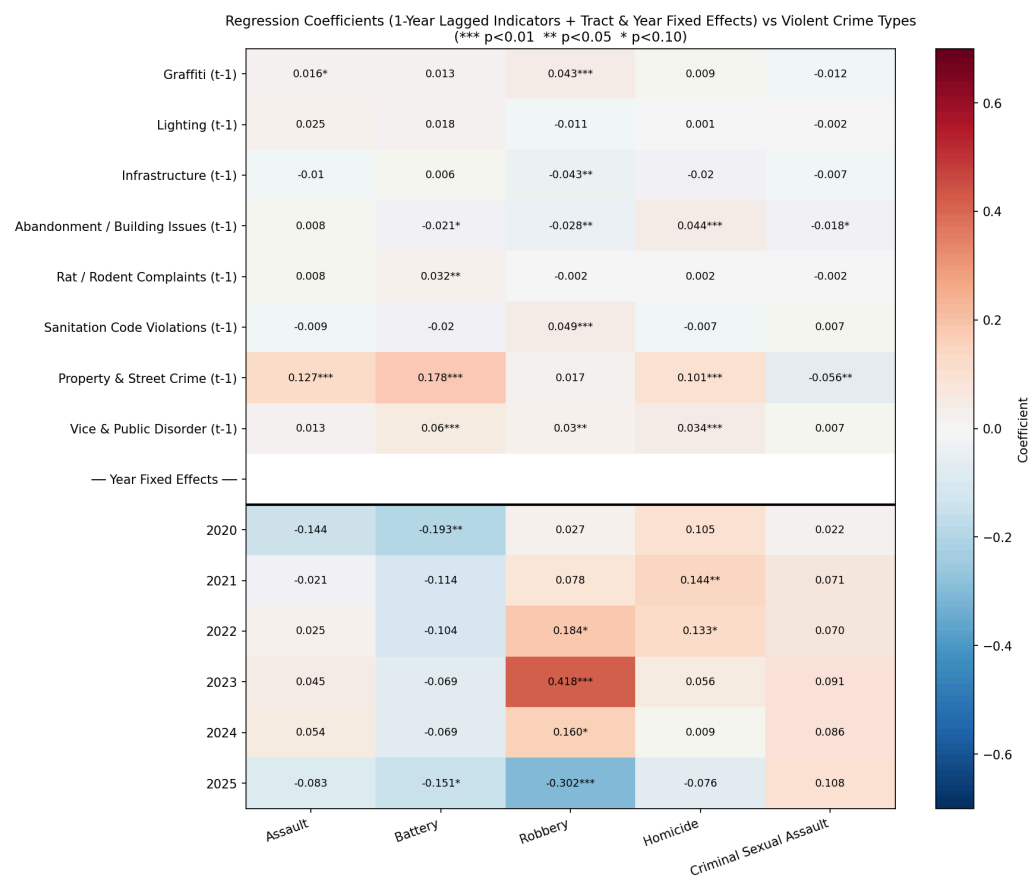


Figure 6 - Violent Crime Rates by Property and Street Crime Quartile (LA Census Tracts)

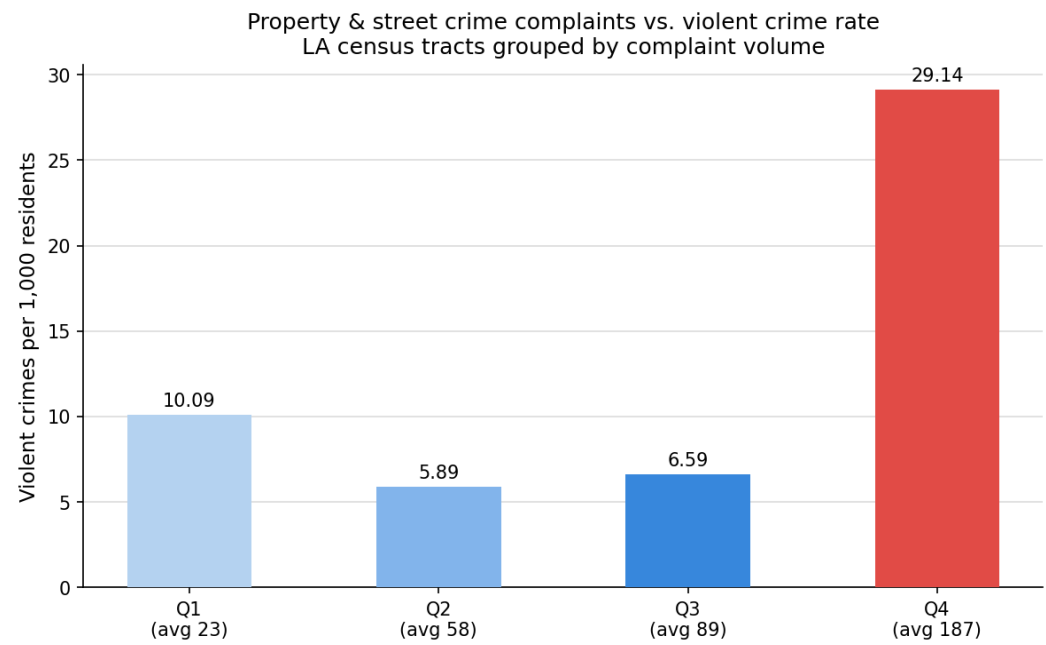


Figure 7 - Spatial Distribution of Homicide Rates by Census Tract

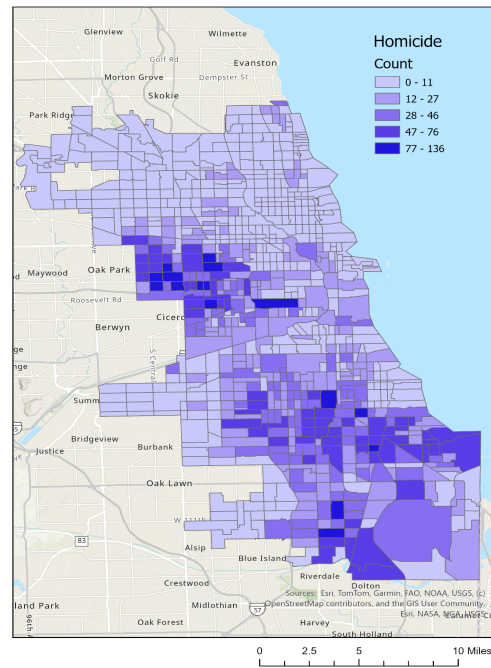


Figure 8 - Vacant and Abandoned Building Complaints by Census Tract

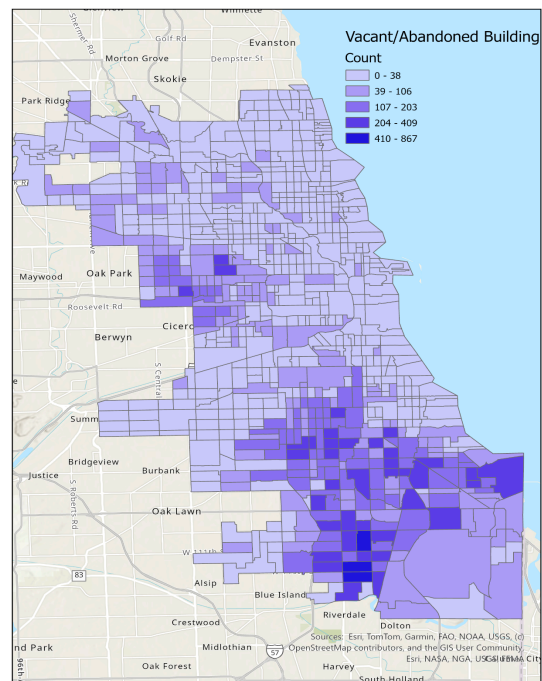


Figure 9 - Spatial Distribution of Assault Incidents by Census Tract

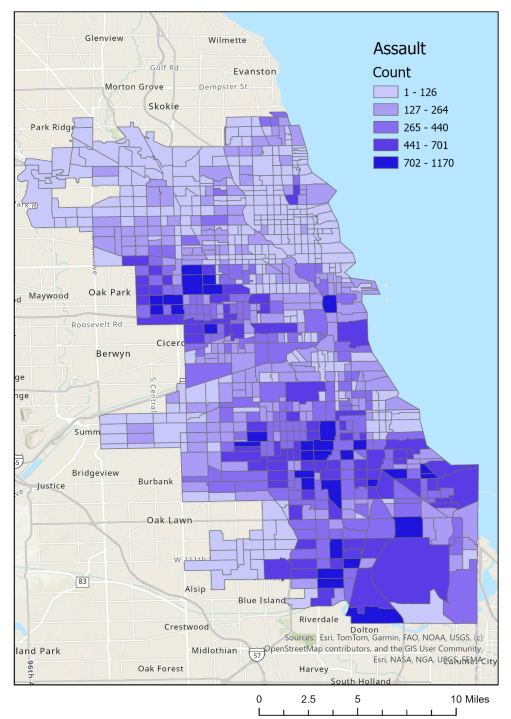


Figure 10 - Spatial Distribution of Robbery Incidents by Census Tract

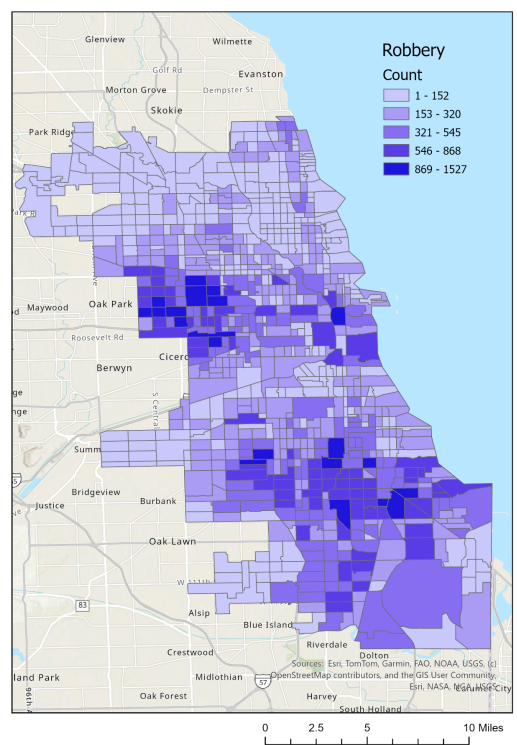


Figure 11 - Spatial Distribution of Battery Incidents by Census Tract

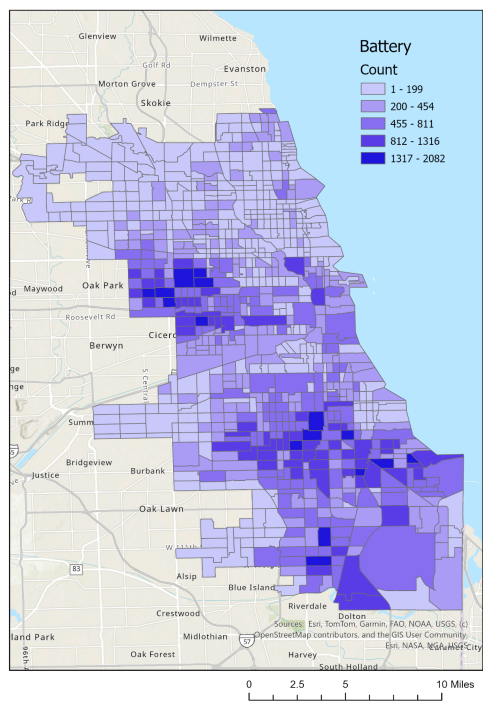


Figure 12 - Spatial Distribution of Criminal Sexual Assault Incidents by Census Tract

